AI-Native Application Modernization Assessment

Assessment Context

As part of this assessment, you will work with an existing application and evaluate how effectively it can be understood and modernized using an AI-assisted engineering approach.

Source repository: https://github.com/mybatis/jpetstore-6

The existing application is the reference point for understanding its current behavior. Interpretations of its behavior and requirements should be grounded in the repository's source code, configuration, documentation, and tests.

Modernization Objective

The objective is to modernize the existing application into a Python-based application while retaining its intended business behavior and addressing appropriate standards for software quality, maintainability, security, reliability, and testability.

Areas of Evaluation

- **Architecture analysis** — understanding the existing application's structure, components, dependencies, and runtime characteristics.
- **Context building** — establishing accurate, useful, and reusable context about the codebase and its behavior.
- **Specification creation** — clarifying the functional and technical requirements relevant to modernization.
- **Architectural decisions** — evaluating the suitability of the target architecture and the trade-offs behind key technical choices.
- **AI guardrails** — considering the constraints, verification practices, and controls needed for responsible AI-assisted engineering.
- **CI/CD** — considering automated build, testing, quality checks, and delivery practices.
- **Demonstration** — providing a live demonstration of the modernized application built during the assessment, showing it running and illustrating its key functionality. The demonstration should also communicate the engineering approach and relevant validation evidence.
- **Tokenomics** — examining AI token usage and resource efficiency in relation to engineering quality and outcomes.

Intended Outcome

The assessment is intended to provide evidence of an engineer's ability to understand an existing application, develop reliable engineering context, make reasoned modernization decisions, and assess the quality of an AI-assisted modernization outcome.

The final outcome should include a working modernized application that the candidate can demonstrate live. Evaluation will consider functional behavior, technical quality, the reasoning behind engineering decisions, the credibility of validation evidence, and the relationship between AI resource consumption and the resulting outcome.